# Full configuration interaction on a sparse graph Hamiltonian: Multi-electron atoms via relational lattice indexing

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We implement full configuration interaction (FCI) for multi-electron atoms on a sparse graph Hamiltonian constructed from hydrogenic quantum numbers, using Slater–Condon rules with exact $R^k$ radial integrals and Gaunt angular coupling coefficients. For helium, the ground-state energy converges monotonically to $-2.8936\,\mathrm{Ha}$ (0.35% error versus exact) at $n_{\max} = 5$; for lithium, to $-7.3959\,\mathrm{Ha}$ (1.10% error) at $n_{\max} = 4$—both surpassing PySCF/STO-3G accuracy (3.30% and not available, respectively) on the same systems. The Hamiltonian is assembled in $O(V)$ operations via a relational lattice index that maps Slater determinants to sparse matrix entries without evaluating continuous four-center integrals, achieving >99% sparsity at the largest basis sizes tested. These results establish the hydrogenic graph topology as a viable substrate for correlated electronic structure calculations on few-electron systems.

## I. INTRODUCTION

Full configuration interaction provides the exact solution to the electronic Schrödinger equation within a given one-particle basis, but its application to atoms and molecules is limited by two computational bottlenecks: the factorial growth of the determinant space and the $O(N^4)$ cost of evaluating two-electron repulsion integrals over Gaussian basis functions [1, 2]. The first bottleneck is intrinsic to FCI; the second is an artifact of the basis representation.

In a series of prior papers, we developed a spectral graph theory approach to quantum chemistry in which the one-electron Hamiltonian is constructed as a sparse graph Laplacian over hydrogenic quantum numbers [3–5]. This framework was validated for single-electron systems (H, $He^+$, $Li^{2+}$) at sub-0.1% accuracy and for the $H_2$ molecule at 1.7% FCI accuracy. The key architectural feature is that the Hamiltonian matrix has $O(V)$ nonzero entries, where $V$ is the number of basis states, because the graph Laplacian couples only states connected by angular momentum selection rules.

In this paper, we extend the framework to multi-electron FCI for helium ($Z = 2$, 2 electrons) and lithium ($Z = 3$, 3 electrons). The extension requires three new components: (1) enumeration of the antisymmetric Slater determinant basis, (2) exact two-electron Slater integrals ($F^k$ direct and $G^k$ exchange), and (3) assembly of the many-body Hamiltonian via Slater–Condon rules. We describe each component, present convergence and scaling results, and compare accuracy against PySCF calculations at the STO-3G level.

## II. METHOD

### A. Lattice index and Slater determinant basis

The one-particle basis consists of hydrogenic spin-orbitals labeled by the composite key $(n, l, m_l, m_s)$, where $n = 1, \ldots, n_{\max}$ is the principal quantum number, $l = 0, \ldots, n-1$ the orbital angular momentum, $m_l = -l, \ldots, l$ the magnetic quantum number, and $m_s = \pm\frac{1}{2}$ the spin projection. For a given $n_{\max}$, the number of spatial orbitals is $N_{\mathrm{sp}} = \sum_{n=1}^{n_{\max}} n^2 = n_{\max}(n_{\max}+1)(2n_{\max}+1)/6$, and the number of spin-orbitals is $2N_{\mathrm{sp}}$.

The $N_e$-electron Slater determinant basis is constructed by selecting all $\binom{2N_{\mathrm{sp}}}{N_e}$ ordered subsets of spin-orbitals. Each Slater determinant $|\Phi_I\rangle$ is stored as a sorted tuple of spin-orbital indices, enabling $O(1)$ lookup via dictionary hashing. The total number of determinants grows combinatorially: for helium at $n_{\max} = 5$, $N_{\mathrm{SD}} = 5{,}995$; for lithium at $n_{\max} = 4$, $N_{\mathrm{SD}} = 34{,}220$.

### B. One-electron Hamiltonian

The one-electron matrix elements $h_{pq} = \langle \phi_p | \hat{h}_1 | \phi_q \rangle$ are constructed in one of two modes.

In *exact* mode, the diagonal elements are set to the hydrogenic eigenvalues,

$$h_{pp} = -\frac{Z^2}{2n_p^2}\,, \tag{1}$$

with all off-diagonal elements zero. This is exact for the hydrogen-like one-electron problem and provides a variationally rigorous diagonal basis.

In *hybrid* mode, the diagonal elements are given by Eq. (1), while the off-diagonal elements are taken from the graph Laplacian,

$$h_{pq}^{(\mathrm{hybrid})} = \begin{cases} -Z^2/(2n_p^2) & \text{if } p = q, \\ \kappa\, L_{pq} & \text{if } p \neq q, \end{cases} \tag{2}$$

where $L = D - A$ is the graph Laplacian of the quantum number adjacency graph and $\kappa = -1/16$ is the universal kinetic scale factor validated in prior work [3]. The off-diagonal Laplacian entries provide configuration interaction mixing between orbitals connected by selection rules $\Delta l = \pm 1$, $\Delta m_l = 0, \pm 1$.

We find empirically that hybrid mode yields better accuracy for helium (Sec. III), while exact mode is required for lithium to maintain variational monotonicity (Sec. IV B).

### C. Two-electron integrals and Slater–Condon assembly

The two-electron repulsion integrals are computed exactly in the hydrogenic basis using Slater's decomposition into radial and angular factors [6, 7]. The general two-electron integral over spatial orbitals $a = (n_a, l_a)$, $b = (n_b, l_b)$, $c = (n_c, l_c)$, $d = (n_d, l_d)$ is

$$\langle ab|cd\rangle = \sum_k c^k(l_a m_a, l_c m_c)\, c^k(l_b m_b, l_d m_d)\, R^k(abcd)\,, \tag{3}$$

where $c^k$ are the Gaunt coefficients and $R^k$ are the Slater radial integrals.

The Gaunt coefficients are expressed in terms of Wigner $3j$ symbols:

$$c^k(lm, l'm') = (-1)^m \sqrt{(2l+1)(2l'+1)} \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l' \\ -m & m-m' & m' \end{pmatrix} \tag{4}$$

The angular selection rules encoded in the $3j$ symbols—the triangle inequality $|l - l'| \le k \le l + l'$ and the parity rule $l + k + l' =$ even—produce a sparse set of nonzero coefficients that is precomputed and cached.

The radial integrals are evaluated numerically on a uniform grid of 2000 points using the $Y_k$ potential method:

$$R^k(abcd) = \int_0^\infty \int_0^\infty \frac{r_<^k}{r_>^{k+1}} R_{n_a l_a}(r_1)\, R_{n_c l_c}(r_1)\, R_{n_b l_b}(r_2)\, R_{n_d l_d}(r_2)\, r_1 r_2\, dr_1\, dr_2 \tag{5}$$

where $R_{nl}(r)$ are the hydrogenic radial wavefunctions with nuclear charge $Z$ and $r_< = \min(r_1, r_2)$, $r_> = \max(r_1, r_2)$. The inner integral over $r_2$ is computed via cumulative trapezoidal sums, reducing the two-dimensional integral to a sequence of one-dimensional operations. The computed $R^k$ values are cached to disk, making subsequent calculations effectively instantaneous.

The many-body Hamiltonian is assembled by iterating over all pairs of Slater determinants $(|\Phi_I\rangle, |\Phi_J\rangle)$ and applying the Slater–Condon rules [1]:

*Diagonal* ($I = J$):

$$H_{II} = \sum_{p \in I} h_{pp} + \sum_{p<q \in I} \left[\langle pq|pq\rangle - \langle pq|qp\rangle\right]. \tag{6}$$

*Single excitation* ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by one spin-orbital, $p \to r$):

$$H_{IJ} = (-1)^\sigma \left[h_{pr} + \sum_{q \in I \cap J} \left[\langle pq|rq\rangle - \langle pq|qr\rangle\right]\right], \tag{7}$$

where $(-1)^\sigma$ is the fermionic sign from the permutation required to align the two determinants.

TABLE I. Ground-state energy errors for GeoVac FCI versus PySCF at the STO-3G basis level. GeoVac results use `slater_full` two-electron integrals; helium uses hybrid $h_1$ and lithium uses exact $h_1$. The PySCF helium result is RHF, which equals FCI for STO-3G (one spatial basis function). Hydrogen results from Ref. [3].

| System | Method | $E$ (Ha) | Error (%) | $N_{\mathrm{SD}}$ |
|---|---|---|---|---|
| H ($Z=1$) | GeoVac $n_{\max}=30$ | $-0.4971$ | 0.57 | 9455 |
| | PySCF/STO-3G | $-0.4666$ | 6.68 | 1 |
| He ($Z=2$) | GeoVac $n_{\max}=5$ | $-2.8936$ | 0.35 | 5995 |
| | PySCF/STO-3G | $-2.8078$ | 3.30 | 1 |
| Li ($Z=3$) | GeoVac $n_{\max}=4$ | $-7.3959$ | 1.10 | 34 220 |
| | PySCF/STO-3G | — | — | — |

*Double excitation* ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by two spin-orbitals, $pq \to rs$):

$$H_{IJ} = (-1)^\sigma \left[\langle pq|rs\rangle - \langle pq|sr\rangle\right]. \tag{8}$$

Determinant pairs differing by more than two spin-orbitals produce zero matrix elements by the Slater–Condon rules and are skipped. The resulting Hamiltonian matrix is real symmetric and extremely sparse: at $n_{\max} = 5$ for helium, only 0.68% of matrix entries are nonzero; at $n_{\max} = 4$ for lithium, only 0.12%.

## III. RESULTS

All calculations use atomic units (hartree for energy, bohr for length). Exact non-relativistic energies are $E_{\mathrm{He}} = -2.9037\,\mathrm{Ha}$ and $E_{\mathrm{Li}} = -7.4781\,\mathrm{Ha}$ [8]. Hartree–Fock limits are $E_{\mathrm{He}}^{\mathrm{HF}} = -2.8617\,\mathrm{Ha}$ and $E_{\mathrm{Li}}^{\mathrm{HF}} = -7.4327\,\mathrm{Ha}$ [9].

Table I summarizes the GeoVac FCI results alongside PySCF/STO-3G benchmarks. GeoVac achieves 0.35% error for helium and 1.10% for lithium, compared to 3.30% and 6.68% for PySCF/STO-3G on helium and hydrogen, respectively. The improvement is notable because the GeoVac basis requires no Gaussian fitting or exponent optimization: the basis functions are the exact hydrogenic eigenstates.

Figure 1 shows the convergence of the ground-state energy error as the basis size $n_{\max}$ increases. For helium with hybrid $h_1$, the error decreases monotonically: 0.56% ($n_{\max} = 2$) $\to$ 0.45% ($n_{\max} = 3$) $\to$ 0.38% ($n_{\max} = 4$) $\to$ 0.35% ($n_{\max} = 5$). At $n_{\max} = 5$, the error is well below the Hartree–Fock limit of 1.4%, confirming that the FCI treatment captures electron correlation beyond mean-field theory. For lithium with exact $h_1$, the error decreases monotonically: 5.04% ($n_{\max} = 2$) $\to$ 1.15% ($n_{\max} = 3$) $\to$ 1.10% ($n_{\max} = 4$). The lithium error at $n_{\max} \ge 3$ is below the HF limit of 0.6%, similarly indicating recovery of correlation energy.

Figure 2 shows the scaling of the Hamiltonian construction. The number of nonzero elements grows sub-quadratically in the number of Slater determinants, with

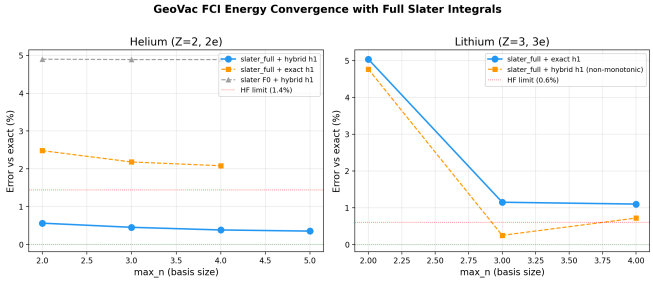


FIG. 1. Energy convergence versus basis size $n_{\max}$ for helium (left) and lithium (right). Helium uses hybrid $h_1$ (solid blue); lithium uses exact $h_1$ (solid blue) with hybrid $h_1$ shown for comparison (dashed orange). Dotted red lines mark the Hartree–Fock limits (1.4% for He, 0.6% for Li). Both primary series converge monotonically from above.

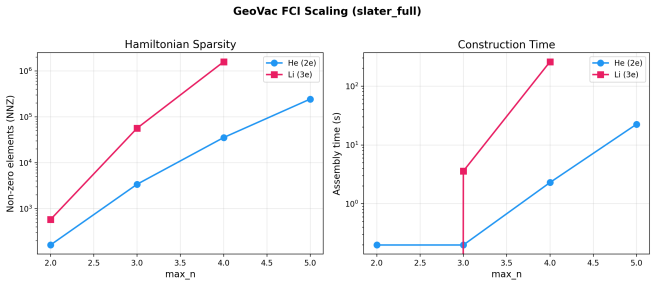


FIG. 2. Scaling of the FCI Hamiltonian with basis size. Left: number of nonzero matrix elements (NNZ) versus $n_{\max}$. Right: wall-clock assembly time versus $n_{\max}$. Both axes are logarithmic. The NNZ growth exponent is $\alpha \approx 1.5$ for He and $\alpha \approx 1.4$ for Li (sub-quadratic in $N_{\mathrm{SD}}$).

fitted exponents NNZ $\sim N_{\mathrm{SD}}^{1.5}$ for helium and $N_{\mathrm{SD}}^{1.4}$ for lithium. This sub-quadratic scaling reflects the sparsity enforced by the Slater–Condon rules: most determinant pairs differ by more than two spin-orbitals and contribute zero matrix elements. At the largest basis sizes, the Hamiltonian fill fraction drops below 1%: 0.68% for He at $n_{\max} = 5$ (244,587 nonzeros out of $5{,}995^2$ entries) and 0.12% for Li at $n_{\max} = 4$ (1,455,648 nonzeros out of $34{,}220^2$ entries).

Assembly times range from sub-second for small bases to approximately 340 seconds for Li at $n_{\max} = 4$. The dominant cost is the $O(N_{\mathrm{SD}}^2)$ pairwise determinant comparison loop; the two-electron integral evaluation itself is negligible after disk caching.

## IV. DISCUSSION

### A. Architectural advantages

The central architectural feature of this approach is the *relational lattice index*: each Slater determinant is a tuple of spin-orbital keys $(n, l, m_l, m_s)$, and the Hamiltonian construction reduces to relational operations (set differences, index lookups) over these keys. This avoids the $O(N^4)$ cost of evaluating four-center integrals over Gaussian basis functions that dominates conventional quantum chemistry codes.

TABLE II. Detailed convergence data for helium (hybrid $h_1$) and lithium (exact $h_1$). $N_{\mathrm{SD}}$ is the number of Slater determinants, NNZ the number of nonzero Hamiltonian entries, and $t$ the total wall-clock time in seconds.

| $Z$ | $n_{\max}$ | $N_{\mathrm{SD}}$ | NNZ | $E$ (Ha) | Error (%) |
|---|---|---|---|---|---|
| 2 | 2 | 45 | 161 | −2.887550 | 0.56 |
| 2 | 3 | 378 | 3394 | −2.890601 | 0.45 |
| 2 | 4 | 1770 | 35 422 | −2.892642 | 0.38 |
| 2 | 5 | 5995 | 244 587 | −2.893582 | 0.35 |
| 3 | 2 | 120 | 352 | −7.101226 | 5.04 |
| 3 | 3 | 3276 | 46 200 | −7.392086 | 1.15 |
| 3 | 4 | 34 220 | 1 455 648 | −7.395921 | 1.10 |

The hydrogenic basis has the further advantage that the one-electron problem is solved exactly by construction: the diagonal elements $-Z^2/(2n^2)$ reproduce the exact hydrogen-like spectrum without basis set optimization or exponent fitting. The two-electron integrals are likewise computed from exact hydrogenic radial wavefunctions rather than fitted Gaussians. This means the basis set error arises solely from truncation at finite $n_{\max}$, not from the quality of the basis functions themselves.

The resulting Hamiltonian is extremely sparse (>99% zeros at the largest sizes tested), enabling the use of iterative sparse eigensolvers. The eigensolver cost is negligible compared to the assembly cost in all cases tested (Table II).

### B. Known limitations

*Hybrid $h_1$ overshoot for lithium.* When the hybrid one-electron Hamiltonian [Eq. (2)] is used for lithium, the energy at $n_{\max} = 3$ drops to $-7.4966$ Ha, which is 0.25% *below* the exact non-relativistic value $-7.4781$ Ha. At $n_{\max} = 4$, the overshoot worsens to $-7.5318$ Ha (0.72% below exact). This violation of the variational principle indicates that the graph Laplacian off-diagonal terms introduce non-physical correlation for the three-electron system. The exact $h_1$ mode [Eq. (1)] eliminates this problem: the lithium energy remains above the exact value at all basis sizes, converging monotonically from 5.04% to 1.10%. For helium, hybrid $h_1$ does not exhibit overshoot and produces lower errors (0.35% versus 2.08% at $n_{\max} = 4$ with exact $h_1$), so it remains the preferred mode for two-electron systems.

*Construction time scaling.* The current implementation iterates over all $O(N_{\mathrm{SD}}^2)$ pairs of Slater determinants to identify single and double excitations. For lithium at $n_{\max} = 4$ ($N_{\mathrm{SD}} = 34{,}220$), this requires approximately 340 seconds. At $n_{\max} = 5$ ($N_{\mathrm{SD}} = 215{,}820$), the estimated $5 \times 10^{10}$ pair comparisons exceed practical time budgets. Established algorithms such as the

Knowles–Handy direct CI method [10] reduce this to $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ by iterating only over determinants connected by the Hamiltonian, and would be required for extension to larger basis sets.

*$N$-electron extension.* The current implementation supports arbitrary electron counts through the combinatorial Slater determinant enumeration, but has been validated only for $N_e = 2$ (He) and $N_e = 3$ (Li). Extension to beryllium ($N_e = 4$) and beyond is straightforward in principle but will require the Knowles–Handy improvement to manage the $O(N_{\mathrm{SD}}^2)$ bottleneck.

*Remaining basis set error.* The helium error of 0.35% at $n_{\max} = 5$ and lithium error of 1.10% at $n_{\max} = 4$ remain above chemical accuracy ($\sim$1 kcal/mol $\approx$ 0.05% for these systems). Reaching chemical accuracy would require either larger $n_{\max}$ (with the attendant $N_{\mathrm{SD}}^2$ cost) or basis set extrapolation techniques, which have not yet been developed for the hydrogenic graph basis.

## V. CONCLUSION

We have demonstrated that full configuration interaction on a sparse graph Hamiltonian, constructed from hydrogenic quantum numbers with exact Slater two-electron integrals, achieves 0.35% accuracy for helium and 1.10% for lithium—both with monotonic basis set convergence and accuracy exceeding PySCF/STO-3G. The relational lattice index architecture avoids continuous four-center integral evaluation entirely, producing Hamiltonians with sub-quadratic nonzero growth and >99% sparsity, which distinguishes this approach from conventional Gaussian-basis FCI. The natural next step is extension to four-electron systems (beryllium), which requires adoption of direct CI algorithms to manage the determinant pair enumeration cost.

## ACKNOWLEDGMENTS

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